

KARL HILL

AEROSPACE & NATIONAL SECURITY SOFTWARE
ENGINEERING SYSTEMS • TECHNICAL LEADERSHIP

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SUMMARY

Staff aerospace software engineer with 20+ years building and delivering production software across national security, aerospace, NASA, and enterprise environments. Hands-on in the code and responsible for the engineering system around it—leading software delivery, raising engineering standards, mentoring engineers, resolving cross-team technical problems, and turning mission needs into reliable software.

PROFESSIONAL EXPERIENCE

Staff Aerospace Software Engineer

SEPT 2025 — PRESENT

Jacobs — National Security • Chantilly, VA

- Lead day-to-day engineering and software delivery for a ~10-engineer team across ~20 repositories and multiple environments: priorities, sprint execution, dependencies, and release readiness.
- Coordinate engineering work across the team and vendors—working their sprint cadences, creating and assigning tickets, implementing and delegating work, and clearing dependencies.
- Establish engineering guardrails in CI/CD, automated testing, code quality, security scanning, repository standards, pull requests, and release controls.
- Anticipate technical, integration, and delivery risks; convene the right engineers and stakeholders and drive issues to resolution before they become blockers.
- Design, implement, and review mission software and shared systems—including broker-independent messaging adapters and post-processing pipeline work—while guiding technical approach across services.
- Onboarded and coached six engineers, reinforcing technical expectations, ownership, and consistent engineering practice.

Lead Software Engineer

DEC 2017 — SEPT 2025

SSAI / NASA Goddard Space Flight Center • Greenbelt, MD

- Led design and development of an AWS-based flood-mapping system, automating processing and distribution of satellite-derived flood products.
- Modernized LAADS DAAC web systems, including Find Data search and ordering and near-real-time access, with GitLab CI/CD and Kubernetes delivery.
- Modernized NASA Earth Observatory's web platform, improving performance, user experience, and search discoverability for an audience of approximately 1.5 million monthly visitors during that work.
- Automated a content-registry workflow, reducing manual dataset-registration work by roughly 60% and improving researcher access.
- Built a Ceph-based file and metadata platform for virtual directory mapping and discovery of large scientific datasets.
- Led Agile software delivery across NASA teams, coordinating priorities, sprint execution, stakeholder needs, and continuous improvement.

DETAILS

Washington, DC

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LINKS

LinkedIn

linkedin.com/in/khill

GitHub

github.com/karlhillx

Website

karlhill.com

AREAS OF EXPERTISE

- Software Engineering & Technical Leadership
- Mission Software & Distributed Systems
- Engineering Systems & Developer Experience
- CI/CD, Testing & Release Engineering
- Systems Integration & Cross-Team Delivery
- Mentoring, Agile Delivery & Engineering Standards

PROFESSIONAL EXPERIENCE

Senior Software Engineer

JAN 2016 — DEC 2017

InformedDNA · Washington, DC

- Designed and delivered a Laravel case-management platform connecting counseling workflows, documentation, and billing.
- Built CRM enhancements for customer lifecycle workflows and reporting, and tightened maintenance and security processes with operations.

Senior Software Engineer

JUN 2012 — MAR 2015

Ticomix, Inc. · Washington, DC

- Delivered SugarCRM solutions for more than 20 clients, including the Virginia Department of Transportation and Kastle Systems.
- Improved backlog management and delivery practices across client projects.

Earlier Engineering Experience

1997 — 2012

Sabre Corporation · Dante Inc. · Visitar Inc. · Verizon Business

- Held software engineering and principal-level roles across travel, enterprise CRM, telecommunications, and managed security.
- Built Java and SQL Server services for Finium, a multi-tenant managed-security platform, and contributed to shared testing and code-quality practices.

AREAS OF EXPERTISE

Software Engineering & Technical Leadership · Mission Software & Distributed Systems · Engineering Systems & Developer Experience · CI/CD, Testing & Release Engineering · Systems Integration & Cross-Team Delivery · Mentoring, Agile Delivery & Engineering Standards

EDUCATION

Computer Science coursework, University of Maryland, Baltimore County

Associate of Arts, General Studies — Computer Science Curriculum, Howard Community College

Project Management studies, Rutgers University

CERTIFICATIONS

SAFe Agilist, Scaled Agile (in progress)

Professional Scrum Master II, Scrum.org

Professional Scrum Product Owner I, Scrum.org

OPEN SOURCE

bb-run — Run Bitbucket Pipelines locally.

testrisk — Rank the highest-value Python test gaps from coverage, AST, and git.

TECHNICAL EXPERTISE

Languages: Python, TypeScript, Java, PHP, Perl, Bash

Cloud & Containers: AWS, Docker, Kubernetes, Helm

CI/CD & Developer Tooling: Git, Bitbucket Pipelines, GitLab CI, GitHub Actions, release automation, security scanning

Python Tooling: uv, Ruff, pytest, mypy, pre-commit

Integration: REST/OpenAPI, RabbitMQ, ActiveMQ, distributed systems

Data: PostgreSQL, MySQL, SQL Server, Elasticsearch/OpenSearch, Redis